Tanay Das

Analog & Mixed-Signal IC Design | ESD Protection | Neuromorphic Computing | RTL Verification | AI-Assisted EDA

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EDUCATION

Indian Institute of Technology, Gandhinagar Gujarat, India
M.Tech, Microelectronics & VLSI, Electrical Engineering Jul 2023 – Jun 2025
CPI: **9.06** *Advisor: Prof. Sandip Lashkare* *Thesis: ESD Protection / Clamping Circuits for Brain-Machine Interface (BMI)*

Sikkim Manipal Institute of Technology Sikkim, India
B.Tech, Electronics & Communication Engineering Jul 2019 – Jun 2023
CPI: **9.7**

RESEARCH INTERESTS

Neuromorphic Computing | Emerging Non-Volatile Memories (RRAM, ReRAM) | ESD Protection for Biomedical SoCs | Analog & Mixed-Signal IC Design | In-Memory Computing | High-Voltage CMOS Design | AI-Assisted EDA Tooling

INDUSTRY EXPERIENCE

AMD (Advanced Micro Devices), Hyderabad Jul 2025 – Present
Silicon Design Engineer — DSV Simulation Team

AI-Assisted EDA Innovation (Production-Deployed, Claude API & Python)

- **AI-Powered RTL Verification Platform**: Full-stack platform auto-generating VHDL/SystemVerilog testcase suites (7 categories, 5 complexity levels) grounded in IEEE LRMs (1076-2008, 1076-2019, 1800-2017) via Claude Sonnet 4.6 + ChromaDB+LlamaIndex RAG. Parallel execution across 4 simulators for instant cross-simulator divergence detection. **5-round auto-fix**, one-click bug filing, WebSocket live streaming, PDF/CSV export.
- **CR Batch Verification Portal**: AI interprets Steps-to-Reproduce, auto-generates simulation scripts, submits to HPC cluster. **18× faster**: 18 bugs in ∼10 min vs. ∼3 hr manually. Live-streamed logs and HTML/CSV reports.

LRM Compliance & Cross-Simulator Verification (XSim, Xcelium, Questa, Riviera-PRO)

- **VIVADO-22278** (VHDL-2019): Built testplan and validated VHDL-2019 constructs — impure functions in port maps, generic subprograms, anonymous types — verifying cross-simulator LRM compliance per IEEE 1076-2019.
- **VIVADO-22280** ($deposit / Signal Forcing): Developed and validated SystemVerilog $deposit signal-forcing flows for Verilog/CPM6 IP block verification; multi-driver, packed-array, and hierarchical forcing scenarios.
- **VIVADO-19753** (VHDL-2008): Built testplan and validated 16 VHDL-2008 constructs — external names, package declarations in protected types, entities, architectures, processes; conditional force assignment.

StarIC, Canada May 2024 – Jul 2024
Photonics IC Design Intern

- Designed WDM EOM and DCF models for a 4-bit serial DAC; layout and simulation using gdsfactory and Python.

Space Application Centre (SAC), ISRO, Ahmedabad Jan 2023 – May 2023
Research Intern

- Designed layout of radiation-hardened digital cell library in 65 nm CMOS using Cadence Virtuoso.
- Implemented EDAC logic; achieved 90% reduction in soft-error rates against single-event upsets.

M.TECH THESIS

ESD Protection / Clamping Circuits for Brain-Machine Interface (BMI)
IIT Gandhinagar *Jan 2024 – Jun 2025* *Advisor: Prof. Sandip Lashkare*

- Designed ESD clamp circuits in 65 nm & 180 nm CMOS; achieved **sub-3 Ω** dynamic resistance supporting 10 V neurostimulator operation, compliant with IEC 61000-4-2.
- Validated via TLP (Transmission Line Pulse) characterization; circuit-level and layout co-design across both technology nodes.
- Resulted in 1 IEEE journal paper (JEDS 2025) and 4 conference papers (EDTM 2025, ICEE 2025 ×2, VDAT 2024).

PUBLICATIONS

Journal *(IEEE Q1 Journal)*

1. **T. Das**, N. Ahmad, L. Somappa, S. Lashkare. "ESD Protection with Sub-3 Ω Dynamic Resistance for Neuromodulation SoC." ***IEEE Journal of the Electron Devices Society (JEDS)***, 2025. [Link]

Conference / Poster *(IEEE international: EDTM, ESSERC; national: ICEE, VDAT)*

1. **T. Das** et al. "IMC Side-Channel Analysis." ***IEEE ESSERC*** (Palma de Mallorca, Spain), Sep. 2026. *(IEEE SSCS Code-a-Chip Travel Grant, VLSI 2026)* [Notebook]
2. **T. Das**, N. Ahmad, L. Somappa, S. Lashkare. "Enhanced ESD Protection for 10 V Neurostimulator Circuits in 65 nm CMOS." ***IEEE EDTM*** (Hong Kong), 2025. [Link]
3. **T. Das**, L. Somappa, S. Lashkare. "ESD Protection with On-chip Diodes for 65 nm CMOS High Voltage Neurostimulators." *ICEE*, 2025. [Link]
4. U. Singh, **T. Das** et al. "Hybrid SW-FPGA RRAM-crossbar for In-Memory Computing with Bit-Slicing." *ICEE*, 2025. **(Best Paper)** [Link]
5. N. Maheshwari, **T. Das** et al. "TCAD-Guided Design of a Compact ESD Protection Circuit for Neurostimulators in 65 nm CMOS." *ICEE*, 2025. [Link]
6. **T. Das**, M. Pathak, S. Lashkare. "Dynamic Resistance Reduction for Voltage Clamp Lowering to Enhance GGNMOS ESD." *VDAT*, 2024. [Link]
7. A. Ghosh, H. Pal, **T. Das**, S. Chatterjee, A. Das. "Synthesis and Characterization of MXene from MAX Phase." *Materials Today: Proceedings*, 2022. [Link]

COURSE PROJECTS

Non-Volatile SRAM Based on ReRAM Technology Aug – Nov 2023
Guide: Prof. Sandip Lashkare, IIT Gandhinagar

- Developed a hybrid memory cell leveraging ReRAM for non-volatility with SRAM-level read/write speed; device modeling and simulation in Verilog-A.

Robust 12-T Interlocked SRAM for In-Memory Computing (IMC) Aug – Nov 2023
Guide: Prof. Joycee Mekie, IIT Gandhinagar

- Designed a 12T SRAM cell with cross-coupled interlocking to enhance read/write stability under IMC workloads; schematic and layout in Cadence Virtuoso.

Analog IC Design Projects: 2-Stage OPAMP, Folded Cascode OTA, Bandgap Reference Jan – Apr 2024
Instructor: Prof. Nihar Ranjan Mahapatra, IIT Gandhinagar

- **2-Stage OPAMP (Miller Comp.)**: Gain >80 dB, CMRR >100 dB, ICMR: 0.8–1.8 V, GBW >10 MHz, PM >65°, Power <0.5 mW.
- **Folded Cascode OTA**: Gain >90 dB, CMRR >100 dB, GBW >10 MHz, Power <20 μW, Output Swing 800 mV.
- **Bandgap Reference**: Temp. regulation ≤10 mV, Supply regulation ≤10 mV, Power <50 μW.

TEACHING EXPERIENCE

- **EE322 — Analog and Mixed Signal Circuits**, IIT Gandhinagar — Tutor, 2024–25 Sem 1
- **ES204 — Digital Systems**, IIT Gandhinagar — Tutor, 2023–24 Sem 2

TECHNICAL SKILLS

Analog Design & Simulation: Cadence Virtuoso (Schematic, Layout, DRC/LVS), NGSpice, Magic, gdsfactory
Digital / HDL: SystemVerilog, Verilog, VHDL, $deposit / signal forcing, LRM compliance testplanning
Simulators: Xilinx XSim, Cadence Xcelium, Mentor Questa, Aldec Riviera-PRO, Xilinx Vivado
AI / Backend: Claude API (Sonnet 4.6), ChromaDB, LlamaIndex, FastAPI, WebSocket, LSF/HPC cluster
Programming: Python (automation, REST APIs, report generation), MATLAB, LaTeX

AWARDS & RECOGNITIONS

- **IEEE SSCS Code-a-Chip Travel Grant** — "IMC Side-Channel Analysis", *IEEE VLSI Symposium 2026*; upcoming poster at **ESSERC 2026** (Sep. 2026) 2026
- **Best Paper Award** — "Hybrid SW-FPGA RRAM-crossbar for IMC", *ICEE 2025* 2025
- **URAM 2.0 Scholarship**, Micron 2024
- **Winner** — "Capture The Bug: Design Verification Hackathon", NIELIT 2022
- **Winner** — "Sky 130 Mixed-Signal Design Marathon", FOSSEE IIT Bombay 2022
- **Gold Medalist** — Certificate of Academic Excellence, SMIT 2021, 2022